



Qualification Pack



Soil Testing Junior Technician

Options: Entrepreneurship management and skills for soil testing

QP Code: DYMT/AGR/Q0101

Version: 1.0

NSQF Level: 2.5

Divya Yog Mandir Trust || Kripalu Bagh, Kankhal, Haridwar
Uttarakhand-249408 || email: vedpriya.arya@patanjali.res.in



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DYMT/AGR/Q0101: Soil Testing Junior Technician

Brief Job Description

Soil Testing Junior Technician

Personal Attributes

Knowledge, Communication, Observations, Technical Skill, Safety responsibilities related to this qualification

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [DYMT/AGR/N0101: Understanding Soil and its Physical Characteristics](#)
2. [DYMT/AGR/N0102: Understanding Nutrient Detection in Soil and Fertilizer Recommendation](#)
3. [DYMT/AGR/N0103: Analyzing soil parameters using Dharti ka Doctor Technology by Patanjali](#)
4. [DYMT/AGR/N0104: Introduction to Organic Methods of Nutrient Management](#)
5. [DYMT/AGR/N0105: Maintaining health and safety guidelines](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Options(Not mandatory):

Option : Entrepreneurship management and skills for soil testing

This unit is about the Entrepreneurship management and skills for soil testing

1. [DYMT/AGR/N0106: Entrepreneurship management and skills for soil testing](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Agriculture Industries
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil



Qualification Pack

Country	India
NSQF Level	2.5
Credits	14
Aligned to NCO/ISCO/ISIC Code	Aligned to NCO/ISCO Code/s 3111.0200 (Laboratory Assistant, Soil)
Minimum Educational Qualification & Experience	9th grade pass OR 8th grade pass with 1.5 years of experience relevant experience (Agriculture and allied sectors OR 5th grade pass with 4.5 years of experience relevant experience (Agriculture and allied sectors OR Ability to read and write with 6 Years of experience relevant experience (Agriculture and allied sectors OR Previous relevant Qualification of NSQF Level 2 with 6 Months of experience of relevant experience (Agriculture and allied sectors
Minimum Level of Education for Training in School	9th Class
Pre-Requisite License or Training	Not Required
Minimum Job Entry Age	15 Years
Last Reviewed On	NA
Next Review Date	25/07/2027
NSQF Approval Date	25/07/2024
Version	1.0
Reference code on NQR	QG-2.5-AG-02836-2024-V1-DYMT
NQR Version	028362024



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DYMT/AGR/N0101: Understanding Soil and its Physical Characteristics

Description

This unit is about the understanding of soil and its physical characteristics

Scope

The scope covers the following :

- Explain the role of soil, soil types, and soil profile
- Demonstrate proper soil sampling techniques
- Explain and demonstrate analysis of soil's physical characters

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Role of soil for plant; types of soil found in India; soil profile; procedure and guidelines for collection of soil samples
- PC2.** Soil physical characteristics- composition, soil texture, soil color, water holding capacity, soil bulk density, soil porosity, soil moisture; Other soil properties- soil pH, electrical conductivity, presence of sulphates and chlorides.
- PC3.** Soil sample collection techniques, Analysis of soil physical characteristics

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Importance of soil quality for enhancing farm productivity
- KU2.** Different types of soil found in India such as clay, loam, and sandy soils
- KU3.** Soil profile and their formation processes
- KU4.** Roles and procedures to operate basic equipment used for the soil analysis
- KU5.** Procedures and guidelines for collecting soil samples to assess their quality and properties effectively
- KU6.** Different types of soils
- KU7.** Identify chemicals/reagents required for conducting various soil physical and chemical tests
- KU8.** • Demonstrate the analysis of soil physical characteristics like composition, texture, color, water holding capacity, bulk density, porosity,
• and moisture content
- KU9.** Demonstrate the analysis of soil pH level, electrical conductivity, and the presence of sulfates and chlorides

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** identify various soil types and soil profile
- GS2.** Perform proper soil sampling techniques
- GS3.** Analyze physical characters of soil



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	30	-	20
PC1. Role of soil for plant; types of soil found in India; soil profile; procedure and guidelines for collection of soil samples	25	5	-	5
PC2. Soil physical characteristics- composition, soil texture, soil color, water holding capacity, soil bulk density, soil porosity, soil moisture; Other soil properties- soil pH, electrical conductivity, presence of sulphates and chlorides.	25	5	-	5
PC3. Soil sample collection techniques, Analysis of soil physical characteristics	-	20	-	10
NOS Total	50	30	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0101
NOS Name	Understanding Soil and its Physical Characteristics
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	2.5
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024



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DYMT/AGR/N0102: Understanding Nutrient Detection in Soil and Fertilizer Recommendation

Description

This Unit is about Understanding Nutrient Detection in Soil and Fertilizer Recommendation

Scope

The scope covers the following :

- Explain soil nutrient functions, deficiency symptoms, and common fertilizers
- Explain nutrient testing technologies available in the market.
- Explain the need for optimal fertilizer use and implementation of fertilizer recommendations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Functions of different soil nutrients for the plants, deficiency symptoms of nutrients in plants; introduction to common fertilizers in use; Problems associated with overuse of fertilizers (Health, environment)
- PC2.** Available soil testing kits in market, applications and limitations
- PC3.** Available fertilizer recommendation apps and their limitations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Explain the essential roles of nutrients such as nitrogen, phosphorus, potassium, and micronutrients in plant growth
- KU2.** Explain nutrient deficiency symptoms in plants
- KU3.** Explain major fertilizers used by farmers
- KU4.** Explain environmental and health risks associated with the overuse of fertilizers
- KU5.** Explain about the available soil testing kits
- KU6.** Explain applications and limitations of available fertilizer recommendation apps.
- KU7.** Demonstration of application methods of fertilizers

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Manage soil nutrient, functions, deficiency symptoms, and common fertilizers
- GS2.** Know about the nutrient testing technologies available in the market
- GS3.** Need for optimal fertilizer use and implementation of fertilizer recommendations



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. Functions of different soil nutrients for the plants, deficiency symptoms of nutrients in plants; introduction to common fertilizers in use; Problems associated with overuse of fertilizers (Health, environment)	10	20	-	5
PC2. Available soil testing kits in market, applications and limitations	5	-	-	2.5
PC3. Available fertilizer recommendation apps and their limitations	5	-	-	2.5
NOS Total	20	20	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0102
NOS Name	Understanding Nutrient Detection in Soil and Fertilizer Recommendation
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	2.5
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024



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DYMT/AGR/N0103: Analyzing soil parameters using Dharti ka Doctor Technology by Patanjali

Description

This unit is about Analyzing soil parameters using Dharti ka Doctor Technology by Patanjali

Scope

The scope covers the following :

- Demonstrate use of Dharti Ka Doctor technology for soil analysis and interpret soil test reports
- Suggest fertilizer recommendation for optimal use of fertilizers

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analysis of soil parameters by Dharti ka Doctor (DKD) soil testing kit; Interpretation of soil test report
- PC2.** Fertilizer recommendation based on soil report using Dharti Ka Doctor Lab App
- PC3.** Analysis of soil nutrients by DKD machine and result interpretation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Explain the basics and applicability of Dharti ka Doctor kit
- KU2.** Explain the basics and applicability of Dharti ka Doctor machine
- KU3.** Demonstrate soil nutrient analysis by Dharti ka Doctor (DKD) soil testing kit
- KU4.** Demonstrate soil nutrient analysis by Dharti ka Doctor (DKD) soil testing machine
- KU5.** Interpret soil testing report
- KU6.** Suggest and implement fertilizer recommendation proved by the Dharti ka Doctor (DKD) Lab app

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Use of Dharti Ka Doctor technology for soil analysis and interpret soil test reports
- GS2.** Suggest fertilizer recommendation for optimal use of fertilizers



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	30	-	20
PC1. Analysis of soil parameters by Dharti ka Doctor (DKD) soil testing kit; Interpretation of soil test report	30	10	-	10
PC2. Fertilizer recommendation based on soil report using Dharti Ka Doctor Lab App	10	10	-	5
PC3. Analysis of soil nutrients by DKD machine and result interpretation	10	10	-	5
NOS Total	50	30	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0103
NOS Name	Analyzing soil parameters using Dharti ka Doctor Technology by Patanjali
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	2.5
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024



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DYMT/AGR/N0104: Introduction to Organic Methods of Nutrient Management

Description

This unit is about Organic farming and Nutrient Management

Scope

The scope covers the following :

- Explain the importance of organic management of fields
- Explain the criteria and processes involved in organic certification
- Demonstrate comparison of organic with normal soil based on appearance and characters
- Explain and demonstrate organic methods of nutrient management in the field

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Importance of organic management of field; Organic certification

PC2. Organic methods of nutrient management- compost, green manures, biofertilizers, crop rotation

PC3. Comparison of organic soil with normal soil

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Importance of organic practices for soil health and sustainability

KU2. Organic certification process and its importance

KU3. Explain organic methods of nutrient management (compost, green manures, biofertilizers, crop rotation)

KU4. Comparison of organic soil with normal soil based on appearance and physical characters

KU5. Comparison of organic soil with normal soil based on nutrient composition

KU6. Organic methods in agricultural settings

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Do organic farming

GS2. Perform Soil management

GS3. To maintain nutrient levels in soil



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	30	-	20
PC1. Importance of organic management of field; Organic certification	25	5	-	5
PC2. Organic methods of nutrient management- compost, green manures, biofertilizers, crop rotation	25	5	-	5
PC3. Comparison of organic soil with normal soil	-	20	-	10
NOS Total	50	30	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0104
NOS Name	Introduction to Organic Methods of Nutrient Management
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	2.5
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024



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DYMT/AGR/N0105: Maintaining health and safety guidelines

Description

This unit is about how to maintain personal hygiene and administer emergency procedures

Scope

The scope covers the following :

- Discuss how to maintain personal hygiene and administer emergency procedures
- Demonstrate the tasks to be performed for conducting the hygiene practices and maintain clean and safe workplace

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Maintenance of personal hygiene, sanitation and safety guidelines of the lab

PC2. Knowledge and Understanding of rules and regulations

PC3. General SOP for working in Lab

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Detail the relevant legislation, standards, policies, and procedures at work
- KU2.** Explain the importance of personal hygiene in laboratory settings
- KU3.** Explain safety measures during working in the field including physical strain, chemical exposure, extreme temperature and biological hazards (bacterial infection, fungal infection and insect stings)
- KU4.** Explain sanitation protocols to prevent contamination and ensure experiment reliability
- KU5.** Explain rules and regulations governing laboratory safety.
- KU6.** Explain safety regulations to mitigate risks and maintain a safe working environment conducive to scientific research and experimentation.
- KU7.** Detail the instructions mentioned on the labels of any reagent bottle or instrument.
- KU8.** List the types of Personal Protective Equipment (PPE) required at the workplace and their importance
- KU9.** Demonstrate hygiene practices such as handwashing and proper lab attire
- KU10.** Demonstrate sanitation procedures for equipment and workspaces
- KU11.** Demonstrate mitigation strategies for any risks in the working environment
- KU12.** Demonstrate how to carry out basic safety checks before the operation of all tools, implements, and machinery and report identified hazards to the supervisor
- KU13.** Demonstrate general SOP for working in Lab



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Maintain personal hygiene and administer emergency procedures
- GS2.** Conduct the hygiene practices and maintain clean and safe workplace



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. Maintenance of personal hygiene, sanitation and safety guidelines of the lab	10	5	-	2.5
PC2. Knowledge and Understanding of rules and regulations	10	5	-	2.5
PC3. General SOP for working in Lab	-	10	-	5
NOS Total	20	20	-	10



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National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0105
NOS Name	Maintaining health and safety guidelines
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	1.0
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024



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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

DYMT/AGR/N0106: Entrepreneurship management and skills for soil testing

Description

This unit is about Entrepreneurship management and skills for soil testing

Scope

The scope covers the following :

- On the Job Training Mapped to DYMT/AGR/N0103 and DYMT/AGR/N0104

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Entrepreneurial Finance, Marketing and Social Entrepreneurship; Quality Control and Assurance

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Soil Properties and Testing Procedures
- KU2.** Soil Sampling
- KU3.** Geotagging the samples
- KU4.** Laboratory Tests on Soil Samples
- KU5.** Soil Classification and Interpretation
- KU6.** Soil Nutrient Analysis
- KU7.** Fertilizer Recommendation
- KU8.** Safety and Quality Standards
- KU9.** Equipment and Tools
- KU10.** Effective Communication
- KU11.** Professionalism and Discipline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Entrepreneurship management and skills for soil testing



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	20	-	10
PC1. Entrepreneurial Finance, Marketing and Social Entrepreneurship; Quality Control and Assurance	20	20	-	10
NOS Total	20	20	-	10



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DYMT/AGR/N0106
NOS Name	Entrepreneurship management and skills for soil testing
Sector	Agriculture
Sub-Sector	
Occupation	Laboratory Assistant, Soil, Laboratory Assistant, Soil
NSQF Level	2.5
Credits	2
Version	1.0
Last Reviewed Date	25/07/2024
Next Review Date	25/07/2027
NSQC Clearance Date	25/07/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- 1.Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.



Qualification Pack

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DYMT/AGR/N0101.Understanding Soil and its Physical Characteristics	50	30	0	20	100	20
DYMT/AGR/N0102.Understanding Nutrient Detection in Soil and Fertilizer Recommendation	20	20	0	10	50	20
DYMT/AGR/N0103.Analyzing soil parameters using Dharti ka Doctor Technology by Patanjali	50	30	0	20	100	20
DYMT/AGR/N0104.Introduction to Organic Methods of Nutrient Management	50	30	0	20	100	20
DYMT/AGR/N0105.Maintaining health and safety guidelines	20	20	0	10	50	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	210	160	-	80	450	100

Optional: 1 Entrepreneurship management and skills for soil testing

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
DYMT/AGR/N0106.Entrepreneurship management and skills for soil testing	20	20	0	10	50	0



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	20	20	-	10	50	0



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
DYMT	Divya Yog Mandir Trust



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above.